

Chapter 2 --- Understanding DBMS, RDBMS and SQL

Applications generate and use huge amounts of data. That data has to be stored, retrieved, updated, and protected efficiently. DBMS, RDBMS, and SQL work together to make this possible.

The relationship in one line: A database stores the data, an RDBMS manages relational databases, and SQL is the language used to communicate with the RDBMS.

2.1 What Is a Database?

A database is an organized collection of related data. For example, a college database may store:

- Student details
- Course details
- Attendance
- Examination marks
- Placement information

A database contains the data, but software is required to manage it --- that software is called a DBMS.

2.2 What Is a DBMS?

DBMS stands for Database Management System. It is software that allows you to create, store, retrieve, update, and delete data in a database. A DBMS provides an interface between:

- The user or application
- The stored data

Without a DBMS, applications would need to manage stored data directly.

Responsibilities of a DBMS

A DBMS helps you:

- Create databases
- Store data
- Retrieve required data
- Modify existing data
- Remove unwanted data
- Control access to data
- Protect data from unauthorized users
- Maintain consistency

- Support backup and recovery

Examples of database management systems: MySQL, PostgreSQL, Oracle Database, Microsoft SQL Server, SQLite, MongoDB. These systems do not all store data using the same model --- systems based on the relational model are called RDBMSs.

2.3 What Is an RDBMS?

RDBMS stands for Relational Database Management System. An RDBMS is a type of DBMS that stores data in related tables. Each table contains:

- Columns --- the properties or attributes of the data
- Rows --- the individual records
- Keys --- columns used to identify and connect records

Example --- a Students table

StudentID StudentName Branch

101 Aarav CSE

102 Bhavna ECE

103 Charan CSE

- Students is the table.
- StudentID, StudentName and Branch are columns.
- Each student is represented by one row.
- StudentID can uniquely identify each student.

Relationships Between Tables

Suppose the database also contains a Marks table:

MarkID StudentID Subject Score

1 101 SQL 85

2 102 SQL 78

3 101 Python 90

The StudentID column connects the Students and Marks tables. This relationship lets the database answer questions such as:

- What marks did Aarav receive?
- Which students scored above 80?
- What is the average score of each student?

Important Features of an RDBMS

- Data is organized into tables.
- Every table can have a primary key.

- Tables can be connected using foreign keys.
- Constraints maintain valid data.
- Duplicate and inconsistent data can be reduced.
- Multiple users can work with the database.
- Transactions help maintain data consistency.

Examples of RDBMSs: MySQL, PostgreSQL, Oracle Database, Microsoft SQL Server, SQLite.

2.4 DBMS vs RDBMS

RDBMS is not separate from DBMS --- it is a specialized type of DBMS.
Every RDBMS is a DBMS, but every DBMS is not necessarily an RDBMS.

DBMS RDBMS

General term for software that manages databases
General term for software that manages databases based on the relational model

Can use different data models Organizes data into related tables

Relationships are not required in every model Relationships are established using keys

Structure depends on the database Uses rows, columns, tables and model constraints

Examples include relational and non-relational systems Examples include MySQL, PostgreSQL and Oracle Database

2.5 What Is SQL?

SQL stands for Structured Query Language. SQL is the language used to communicate with a relational database management system. You write SQL commands, and the RDBMS executes them on the database.

SQL can be used to:

- Create databases and tables
- Insert records
- Retrieve records
- Filter and sort data
- Update records
- Delete records
- Combine data from multiple tables

- Calculate totals and averages
- Control user permissions
- Manage transactions

Examples of SQL Commands

Create a table:

```
CREATE TABLE Students ( StudentID INT PRIMARY KEY, StudentName VARCHAR(50), Branch VARCHAR(20));
```

Insert a record:

```
INSERT INTO Students VALUES (101, 'Aarav', 'CSE');
```

Retrieve records:

```
SELECT *FROM Students;
```

Update a record:

```
UPDATE StudentsSET Branch = 'ISE'WHERE StudentID = 101;
```

Delete a record:

```
DELETE FROM StudentsWHERE StudentID = 101;
```

2.6 How They Work Together

Suppose a college wants to manage student data:

- The student information is stored in a database.
- MySQL manages that database, so MySQL is the RDBMS.
- You write SQL commands to create, retrieve, and modify the data.

Term Role in the College Application

Database Contains student, course and marks data

DBMS Software that manages the database

RDBMS Manages data using related tables

MySQL The RDBMS being used

SQL Language used to communicate with MySQL

SQL and MySQL Are Not the Same

- SQL is a language.
- MySQL is an RDBMS that understands SQL.

Other RDBMSs --- such as PostgreSQL, Oracle Database, and SQL Server --- also use SQL, though their syntax may differ slightly for some operations.

A useful comparison: SQL is the language, while MySQL is the software that understands and executes that language.

Common Confusions, Cleared Up

Question Answer

Is a database the same as a DBMS? No. A database contains the data. A DBMS is the software used to manage that data.

Are DBMS and RDBMS opposites? No. RDBMS is one type of DBMS.

Is SQL a database? No. SQL is a language used to work with relational databases.

Is MySQL a programming language? No. MySQL is an RDBMS. It accepts and executes SQL commands.

Can SQL be used with every database? SQL is mainly used with relational databases. Non-relational databases may use different query languages or APIs.

Interview Questions — Chapter 2

Q1. Is every DBMS an RDBMS?

Ideal answer: No. RDBMS is a type of DBMS based on the relational model.

Q2. Is MySQL the same as SQL?

Ideal answer: No. SQL is a language; MySQL is an RDBMS that accepts and executes SQL.

Q3. Why are relationships important in an RDBMS?

Ideal answer: Relationships connect related records across tables using keys, allowing related information to be retrieved without storing everything in one table.

Q4. What is the role of a DBMS?

Ideal answer: It manages operations such as storing, retrieving, updating, deleting, access control, consistency, backup, and recovery.

Q5. Can non-relational databases use SQL?

Ideal answer: Some non-relational systems may provide SQL-like languages or SQL interfaces, but SQL is primarily associated with relational databases.